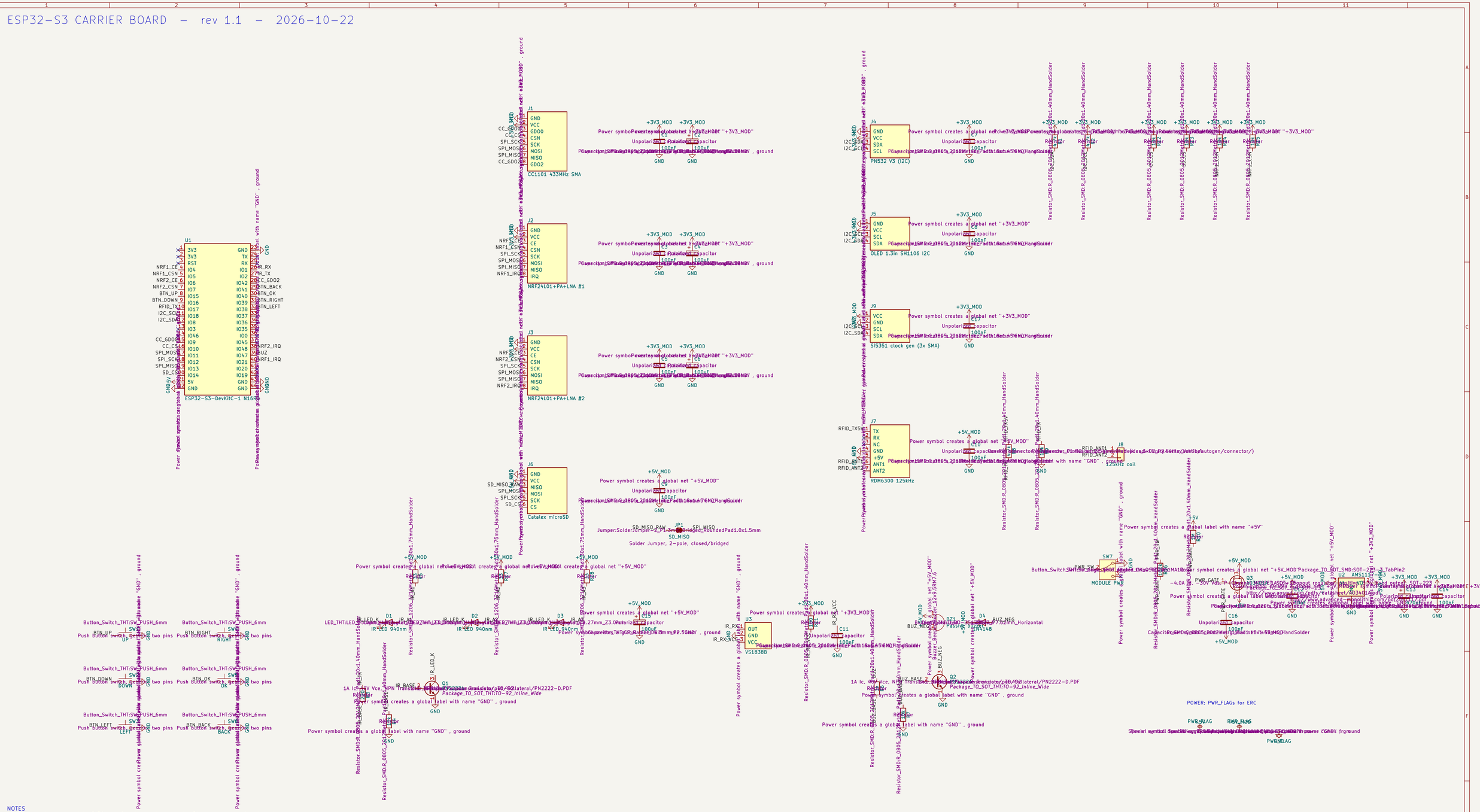

ESP32-S3 CARRIER BOARD – rev 1.1 – 2026-10-22
U1
ESP32-S3-DevKitC-1 N16R8
J1
CC1101 433MHz SMA
J2
NRF24L01+PA+LNA #1
J3
NRF24L01+PA+LNA #2
J6
Catalex microSD
J4
PN532 V3 (I2C)
J5
OLED 1.3in SH1106 I2C
J9
SI5351 clock gen (3x SMA)
J7
RDM6300 125kHz
J8
125kHz coil
JP1
Solder Jumper, 2-pole, closed/bridged
SW7
MODULE PWR
Q3
Q2
U2
AMS1117-3.3
U3
VS1838B
D1
D2
D3
D4
BTN_UP
BTN_DOWN
BTN_LEFT
BTN_RIGHT
BTN_OK
BTN_BACK
NOTES
1) Every module plugs into 2.54 mm female header sockets. Pin names are printed next to each socket on the silkscreen.
2) Power: DevKit 5V pin -> +5V. SW7 (slide) drives Q3 (P-MOSFET) with a soft-start (R16 100k + C16 100nF, ~20 ms ramp): +5V -> +5V_MOD.
+5V_MOD -> microSD, RDM6300, IR LEDs, buzzer and U2 (AMS1117-3.3) -> +3V3_MOD (all 3.3 V modules).
SW7 OFF: the ESP32 keeps running, every module is unpowered. The ESP32's own 3V3 pin is not used (no overload / brown-out).
3) Shared SPI: MOSI=IO11 SCK=IO12 MISO=IO13. CS: CC1101=IO10, SD=IO14, NRF1 CSN=IO5, NRF2 CSN=IO7 (10k pull-ups R12-R15).
CE: NRF1=IO4, NRF2=IO6. CC1101 GDO0=IO9, GDO2=IO42. NRF IRQ: NRF1=IO21, NRF2=IO48 (optional, can be ignored by firmware).
4) Shared I2C: SDA=IO8 SCL=IO18, 4.7k pull-ups to +3V3_MOD (R1, R2). Devices: OLED SH1106 (0x3C), PN532 (0x24), SI5351 (0x60).
PN532 V3 DIP switch: SW1=ON SW2=OFF for I2C.
5) RDM6300 TX is 5 V logic -> R8/R9 divider (~3.4 V) -> IO17. The 125 kHz coil plugs into J8.
6) IR receiver VS1838B on IO1 (R11/C11 supply filter, lens faces the top edge). IR transmitter on IO2: Q1 + 220R base,
three IR LEDs, each with its own 33R resistor (~95 mA peak each). C15 local reservoir. Buzzer on IO47 (Q2 + 1N4148).
7) Buttons pull to GND (enable internal pull-ups): UP=IO15 DOWN=IO16 LEFT=IO38 RIGHT=IO39 OK=IO40 BACK=IO41.
Note: on DevKitC-1 v1.1 IO38 also drives the on-board RGB LED.
8) JP1: normally-closed solder bridge. Old Catalex modules never release MISO; if SPI devices clash, cut JP1 and tie the
module's 74LVC125 MISO OE pin to CS (or add a tri-state buffer here).
9) Unused GPIO: 0, 3, 19, 20, 35–37, 45, 46, TX/RX, RST.
ESP32-S3-DevKitC-1 N16R8 + CC1101 + 2x NRF24 + PN532 + RDM6300 + SI5351 + microSD + O
github.com/Yoursel71
Sheet: /
File: ESP32S3_Carrier.kicad_sch
Title: ESP32-S3 Carrier Board
Size: A2
Date: 2026-10-22
Rev: 1.1
KiCad E.D.A. 10.0.6
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